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High resistance of fluoroquinolone and macrolide reported in avian pathogenic *Escherichia coli* isolates from the humid subtropical regions of Pakistan

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ABSTRACT

Objectives: This study aimed to assess the antimicrobial resistance profile, virulence potential, and genetic characterization of avian pathogenic *Escherichia coli* (APEC) that cause colibacillosis in poultry.

Methods: Antibiotic susceptibility testing (AST) was measured via the Kirby-Bauer disc diffusion method against 27 commonly used antibiotics. Phylogrouping, virulence-associated gene detection, and hybrid strain detection via multiplex polymerase chain reaction (PCR) and genetic diversity were analysed via ERIC-PCR fingerprinting method.

Results: AST analysis showed 100% of isolates were multidrug-resistant (MDR) and highest resistance was against penicillin, tetracycline, and macrolide classes of antibiotics. The *mcr-1* gene was present in 40% of the isolates, though only 4% of isolates were showing phenotypic resistance. Despite the scarce use of fluoroquinolone, carbapenem, and cephalosporin in the poultry sector, resistance was evident because of the high prevalence of extended-spectrum β -lactamase (ESBL) (53.7%) and other β -lactamases in APEC isolates. β -lactamase genotyping of APEC isolates revealed that 85.7% of isolates contained either *blaCTX* or *blaTEM* and around 38% of isolates were complement resistant. Growth in human urine was evident in 67.3% of isolates. Phylogroup B1 (51%) was the most prevalent group followed by phylogroups A (30.6%), D (13.61%), and B2 (4.76%). The most prevalent virulence-associated genes were *fimH*, *iss*, and *tatT*. Results showed that 26 isolates (17.69%) can be termed hybrid strains and APEC/EHEC (enterohemorrhagic *E. coli*) was the most prevalent hybrid *E. coli* pathotype. ERIC-PCR fingerprinting genotype analysis clustered APEC isolates in 40 groups (E1–E40). This study provides insights into the antibiotic resistance and virulence profiling of the APEC isolates in Pakistan.

Conclusions: The findings of this study provide insights into that the antibiotic resistance and virulence profiling of the APEC isolates in Pakistan. This data can inform future studies designed to better estimate the severity of the colibacillosis in poultry farms.

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1. Introduction

Escherichia coli is a normal inhabitant of the gastrointestinal tract of animals, birds, and humans. While often harmless, pathogenic variants (pathotypes) exist, including avian pathogenic *Escherichia coli* (APEC). These pathotypes can invade distinct organs of poultry flocks and cause airsacculitis, pericarditis, perihepatitis,

and peritonitis. If left untreated, APEC infections can lead to septicemia. All these conditions are collectively termed 'colibacillosis', which is a systemic and localized infection [1]. Colibacillosis causes reduction in the weight of chicken by up to 2%, decreases egg production by up to 20%, and has a mortality rate of up to 20% [2]. In young chickens, the mortality rate due to APEC infection can be as high as 53.5% [3], contributing to huge economic losses for the poultry industry worldwide.

Virulence factors that are responsible for the pathogenesis of APEC include colonization factors (responsible for the attachment of APEC to the extraintestinal tract), adhesions (facilitating the

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attachment and penetration of APEC to the extraintestinal tract), toxins (shielding the APEC from lysosomes), and siderophores (iron-chelating agents) [4]. These factors are responsible for the establishment of APEC infection and resistance against the host immune defence system. Virulence genes usually associated with the pathogenic capability of APEC are *iucD*, *iroN*, *tsh*, *vat*, *hlyF*, *iss*, *ompT*, and *cvi/cva*. APEC is identified by the presence of five or more of these virulence genes [5]. Many studies report broad serological diversity among APEC groups [6,7]. Predominant serotypes that are responsible for more than 80% of colibacillosis outbreaks worldwide are O1, O2, and O78 [8,9]; however, they account for only 15% to 60% of the serogroups that cause colibacillosis [10–12].

Epidemiological data show that APEC and human extraintestinal pathogenic *E. coli* (ExPEC) have a similar phylogenetic background and carry overlapping sets of virulence genes, suggesting a zoonotic potential for APEC strains [13]. Sequencing of APEC strain O1:K1:H7 revealed a high nucleotide identity to the genome of human uropathogenic *E. coli* (UPEC) and neonatal meningitis *E. coli* (NMEC) [14]. Moreover, studies report the presence of APEC-specific Colicin V plasmids in human ExPEC strains, suggesting the zoonotic transmission of APEC isolates from poultry to humans [15]. Humans can acquire these pathogens through the faecal-oral route [5]. Suggested zoonotic transmission of APEC makes these pathogens important to the poultry industry as well as public health.

Hybrid *E. coli* strains are those strains that carry both ExPEC and DEC (diarrheagenic *E. coli*) markers. ExPEC cause extraintestinal infections and are distinguished from DEC that cause intestinal infections. However, some ExPEC strains harbour some toxin genes present in DEC; such strains are DEC/ExPEC hybrid strains [16]. The genome of *E. coli* is continuously emerging through horizontal gene transfer crossing pathovar definition. Therefore, the term 'hybrid' is coined to report the new combinations of virulence gene content among classic *E. coli* pathotypes.

Antibiotics, such as aminoglycosides, tetracyclines, and sulfonamides, are widely being used to control colibacillosis in poultry flocks [17]. In developing countries, farmers have been using these antibiotics for many years for multiple purposes, such as for the prevention of disease and growth promotion. Excessive use of antibiotics in poultry flocks have led to detrimental effects, including the emergence of antimicrobial-resistant strains and the presence of antibiotic residues in poultry meat and products [18]. Emerging resistance against antibiotics, including those that are medically important, such as carbapenems, fluoroquinolones, β -lactams, and colistin, propose certain challenges in controlling APEC infections in chickens [19]. Data regarding virulence traits and antibiotic susceptibility patterns among APEC isolates is lacking in Pakistan. Therefore, the present study was designed to fill these gaps.

2. Materials and methods

2.1. Sample collection and *E. coli* isolation

A total of 150 dead broiler chickens infected with colibacillosis were collected from different poultry farms of the humid subtropical regions of Pakistan and examined for the gross lesions' characteristic of colibacillosis in the Veterinary Diagnostic Laboratory located in the animal house of National University of Sciences and Technology (NUST), Pakistan. Liver samples ($n = 150$) showing characteristic fibrinous hepatitis were collected aseptically for APEC isolation. The liver samples (1 g) were ground, mixed with 10 mL of Peptone-buffered water (Oxoid™, UK), and incubated overnight at 37°C in a shaking incubator. One loopful of the incubated sample was streaked onto MacConkey-lactose agar

(Oxoid™) and incubated at 37°C for 24 h. One lactose fermenting colony was picked from the plate and streaked onto Eosin methylene blue (EMB) agar (Oxoid™). One colony with a characteristic green metallic sheen was used for further experimentation. The isolates were confirmed as *E. coli* by polymerase chain reaction (PCR) detection of the *E. coli*-specific housekeeping gene, *uidA* [20]. Boiled DNA extract was used as template for PCR.

2.2. ERIC-PCR fingerprints analysis

To confirm that APEC isolates used in this study were diverse in nature, ERIC-PCR was employed. To ensure reproducibility of the banding pattern, ERIC-PCR was conducted a minimum of two times for each isolate. Results were analysed using the software package BioNumerics 7.5 (Applied Maths, Keistraat, Belgium). The UPGMA (unweighted pair group method with arithmetic mean) method was employed to create a similarity dendrogram with a 1% tolerance limit and 1% optimization using the DICE similarity coefficient. An 80% similarity cut-off was used for identification of the clusters [21].

2.3. Antimicrobial susceptibility testing

Isolated *E. coli* strains were subjected to antibiotic susceptibility testing via the Kirby-Bauer disc diffusion method. Altogether, 26 antibiotics were procured from Oxoid™ (Supplementary File S1). As per the Clinical and Laboratory Standards Institute (CLSI) guidelines [22], the diameters of the zones of inhibition were measured in millimetres and termed as susceptible (S), intermediate (I), or resistant (R). *Escherichia coli* ATCC-25922 was used as an antibiotic-sensitive control. Results from the antibiotic susceptibility tests were visualized via a heat map illustrating the hierarchical clustering of each isolate in the software package BioNumerics 8.0 (Applied Maths, Keistraat, Belgium).

In order to determine minimum inhibitory concentrations (MICs) to colistin, APEC isolates were inoculated in Mueller-Hinton (MH) broth in a range of colistin concentrations spanning from 0.125 $\mu\text{g/mL}$ to 128 $\mu\text{g/mL}$. Following overnight incubation in microtiter plates at 37°C, the lowest colistin concentration that resulted in no visible bacterial growth was identified as the MIC endpoint. Colistin-sensitive and colistin-resistant *E. coli* isolates were distinguished by MICs $\leq 4 \mu\text{g/mL}$ and $> 4 \mu\text{g/mL}$, respectively [23].

2.4. Screening and confirmation of ESBL production

APEC isolates that displayed a diameter ≤ 22 mm for cefazidime and/or ≤ 27 mm for cefotaxime were subjected to ESBL production confirmation via the Double Disc Synergy Test (DDST). Distortion in zone towards AMC or increase in zone of inhibition by 5 mm were considered positive results for ESBL production [22].

2.5. Screening and confirmation of metallo β -lactamase (MBL) production

APEC isolates that exhibited resistance against ceftazidime (CAZ, 30 μg), imipenem (IPM, 10 μg), and/or meropenem (MEM, 10 μg) were subjected to MBL production confirmation testing via an inhibition method whereby EDTA was employed as an inhibitor [24].

2.6. Modified Hodge test for the detection of carbapenemase production

To identify carbapenemase production by APEC isolates, a modified Hodge test was used. A dilution of *E. coli* ATCC 25922 was spread on MHA plates; a 10 μg meropenem disc was placed in each

plate's centre and the corresponding *E. coli* isolate was streaked in a straight line from the disc edge to the plate edge. Following incubation at 37°C for 24 h, a clover leaf-like indentation of the *E. coli* ATCC 25922 growing along with the test APEC isolates growth within the disk diffusion zone was considered a positive result for carbapenemase production [25].

2.7. AmpC disc test for the detection of AmpC β -lactamase production

APEC isolates that yielded a zone diameter <18 mm for cefoxitin (FOX, 30 μ g) were tested for the AmpC enzyme by AmpC disc test. AmpC discs rehydrated with 20 μ L of saline were inoculated with several colonies of a tested *E. coli* isolate and the resultant disc was then placed near the FOX disc. Following incubation at 37°C for 24 h, the plates were examined for any distortion or flattening of the zone of inhibition, indicating the enzymatic inactivation of FOX (a positive result) [26].

2.8. Determination of Multiple Antibiotic Resistance Index (MARI) and Resistance score (R-Score)

The MARI of APEC isolates was estimated to establish their multiple antibiotic resistance profiles [27]. The MARI formula was described as:

MARI = a/b - Equation 1 a = The number of antibiotics to which the tested APEC isolate was resistant to b = The total number of antibiotics to which the tested APEC isolate has been evaluated for

For a given APEC isolate, its R-score was described as the number of antibiotics against which the isolate exhibited intermediate or complete resistance. Resistance scores of 0.5 and 1 were attributed to isolates exhibiting intermediate or complete resistance, respectively, against a given antibiotic [28].

2.9. Detection of antibiotic resistance genes

All APEC isolates were scrutinized for colistin resistance genes through the detection of *mcr-1* and *mcr-2* genes [29]. The APEC isolates were further examined for ESBL genes (*blaCTX*, *blaTEM*, and *blaSHV*), MBL genes (*blaVIM*, *blaNDM*, and *blaIMP*), and carbapenemase genes (*blaKPC* and *blaOXA-48*) by PCR using primers and conditions described by Doyle [30] (Supplementary File S1). *Klebsiella pneumoniae* ATCC 700603 was used as a positive control and *E. coli* ATCC 25922 was used as a negative control for the identification of ESBL genes.

2.10. In vitro pathogenicity analysis

All APEC isolates were subjected to phenotypic analysis for biofilm formation ability, blood hemolysis, congo red (CR) binding ability, complement resistance, growth in human urine, lipase activity, motility (swimming, swarming, and twitching), and protease activity.

Biofilm assay for APEC isolates was performed according to the microtiter plate method [31]. The optical density (OD) of isolates was adjusted at 1.0, diluted 1:100 in Tryptone soy broth (TSB), and the diluted sample was added to the 96-well plate and incubated at 37°C for 48 h. Planktonic cells were removed; loosely bound cells were removed by washing with saline and adherent cells were stained with 0.1% crystal violet (CV) solution. Stained cells were resuspended in 33% glacial acetic acid, and absorbance was recorded at 595 nm. *Bacillus subtilis* was used as a positive control. The biofilm forming ability of each isolate was calculated by the given formula:

ODc: average OD of negative control + 3 $\times$ standard deviation (SD) of negative control

OD $\leq$ ODc – no biofilm production

ODc < OD $\leq$ 2 $\times$ ODc – weak biofilm production

2 $\times$ ODc < OD $\leq$ 4 $\times$ ODc – moderate biofilm production

4 $\times$ ODc < OD – strong biofilm production

Congo red binding assay was conducted in accordance with the method described by Berkhoff [32]. Congo red agar plates were streaked with APEC isolates, incubated for 24 h at 37°C, and left for another 2 days at room temperature. Each isolate was visually observed for binding with CR dye. The CR positive samples appeared red, while CR negative samples appeared as white to pink colonies. For blood hemolysis assay, APEC isolates were streaked on blood agar plates infused with 5% aseptic, fresh, defibrinated sheep blood. To interpret the results, guidelines by Buxton [33] were used.

Growth in human urine was performed to analyse if APEC isolates can grow in the high urea and nitrogenous environment of urine. The assay was performed by the method described by Mitchell et al. in 2015 with little modifications [31,34]. For the collection of urine samples, healthy male and female subjects were asked to sign the form approved by the IRB committee of the Industrial Biotechnology Department of NUST (Ref No: IRB-88). Collected urine samples were filter sterilized (with a 0.2 μ m sterile filter), pooled, and stored in aliquots at –20°C. Briefly, OD of tested isolates was adjusted to 1.0. A 1:100 dilution of isolates was prepared in sterile urine and the dilutions were dispensed in a microtiter plate. The plate was incubated at 37°C for 8 h under static conditions, and absorbance was recorded at 600nm. UPEC strain CFT073 and *E. coli* DH5 α strain were used as positive and negative controls, respectively.

Resistance to foetal bovine serum complement was analysed by quantitative microtiter plate method with slight modification [35]. Briefly, 10⁴ CFU of APEC isolates were mixed with an equal volume of 50% serum and the resultant mixture was added to a 96-well microtiter plate. The plate was incubated at 37°C for 4 h under static conditions and OD₄₉₂ was determined. *Escherichia coli* isolates were considered complement resistant if the OD₄₉₂ in serum-containing wells exceeded or equalled that of the no-serum control well. Heat-inactivated sera was used as a control. UPEC strain CFT073 and *E. coli* K-12 MG1655 strain were used as positive and negative controls, respectively.

Lipase activity of the APEC isolates was conducted by streaking the isolates on tryptic soy agar (TSA) supplemented with 1% Tween 80, followed by incubation for 1 to 2 days at 37°C. Clear halo zones around bacterial growth were considered positive for lipase activity. Protease activity of the APEC isolates was conducted by streaking the isolates on TSA supplemented with 1% casein from bovine milk (Sigma Aldrich, Germany), followed by incubation for 1 to 2 days at 37°C. APEC isolates positive for protease activity were identified by clear halo zones around bacterial growth.

Motility assays investigating the swimming, swarming, and twitching patterns of APEC isolates were conducted by methods described earlier [36]. Freshly poured media plates were dried at room temperature for 6 h. Then, APEC isolates were stabbed on corresponding plates followed by incubation at 37°C for a specific time (8–12 h for swimming and swarming and 24 h for twitching motility). Swimming and swarming motility was analysed by measuring the halo zone around the bacterial stab. Agar media from the twitching motility plates was removed and each plate was stained with 1% CV for 15 minutes. Finally, the stained zone was measured for the twitching motility assay; a minimum of six colonies from each APEC isolate was tested for each motility assay. *Pseudomonas aeruginosa* was used as a positive control for motility assay.

Table 1

Table representing diarrheagenic *E. coli* (DEC) pathotypes along with their detection markers

Sr. No	DEC Pathotype	Detection Marker
1	Typical EPEC	<i>eaeA</i> , <i>bfpB</i>
2	Atypical EPEC	<i>eaeA</i>
3	EHEC	<i>stx1</i> , <i>stx2</i>
4	EAEC	<i>aggR</i>
5	EIEC	<i>invE</i>
6	ETEC	<i>lt</i> , <i>stx</i> , <i>stx</i>

2.11. Phylogenetic classification of APEC isolates

APEC isolates were classified into different phylogenetic groups (phylogroups A, B1, B2, and D) using a single multiplex PCR as described by Clermont [37]. Based on the presence or absence of three DNA markers (*chuA*, *yjaA*, and DNA fragment TSPE4.C2), APEC isolates were assigned to specific phylogenetic groups. The triplex PCR conditions, as well as primer sequences, are mentioned in Supplementary File S1 (Table 2).

2.12. Detection of virulence-associated genes (VAGs)

APEC isolates were screened for various ExPEC-associated virulence genes by multiplex PCR. The primers and their conditions are mentioned in Supplementary File S1. Three replicates were used for each gene identification. UPEC strain CFT073 and some clinical ExPEC strains (provided by Antibacter Lab) were used as positive controls.

2.13. Detection of DEC-related genes

To analyse whether APEC isolates had toxin genes or not, they were examined for the presence of the DEC virulence-encoding genes mentioned in Table 1. Six groups of DEC were analysed, including enterohaemorrhagic *E. coli* (EHEC), enteropathogenic *E. coli* (EPEC), enteroinvasive *E. coli* (EIEC), enterotoxigenic *E. coli* (ETEC),

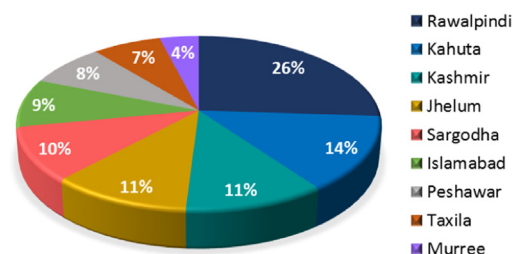
Distribution of APEC isolates by location

Fig. 1. Pie chart showing the distribution of the Avian pathogenic *E. coli* (APEC) isolates according to their isolation locations.

and enteroaggregative *E. coli* (EAEC). The primers and their conditions are mentioned in Supplementary File S1. *Escherichia coli* O157:H7 (ATCC 35150) was used as positive control for *stx1*, *stx2*, and *eaeA* genes.

3. Results

3.1. Distribution of APEC isolates

A total of 147 isolates were confirmed to be *E. coli* by green metallic sheen appearance on EMB agar and *uidA* gene presence. The relative distribution of these 147 APEC isolates by their isolation locations from the humid subtropical regions is summarized in Fig. 1; most of the APEC strains were isolated from Rawalpindi (26%), followed by Kahuta (14%).

3.2. In vitro antibiotic susceptibility pattern of the isolates

All isolates showed resistance towards the macrolide erythromycin. The next highest level of resistance was for the penicillins, as 136 isolates (92.5%) were resistant to both ampicillin and amoxicillin. Around 80.9% isolates were resistant to

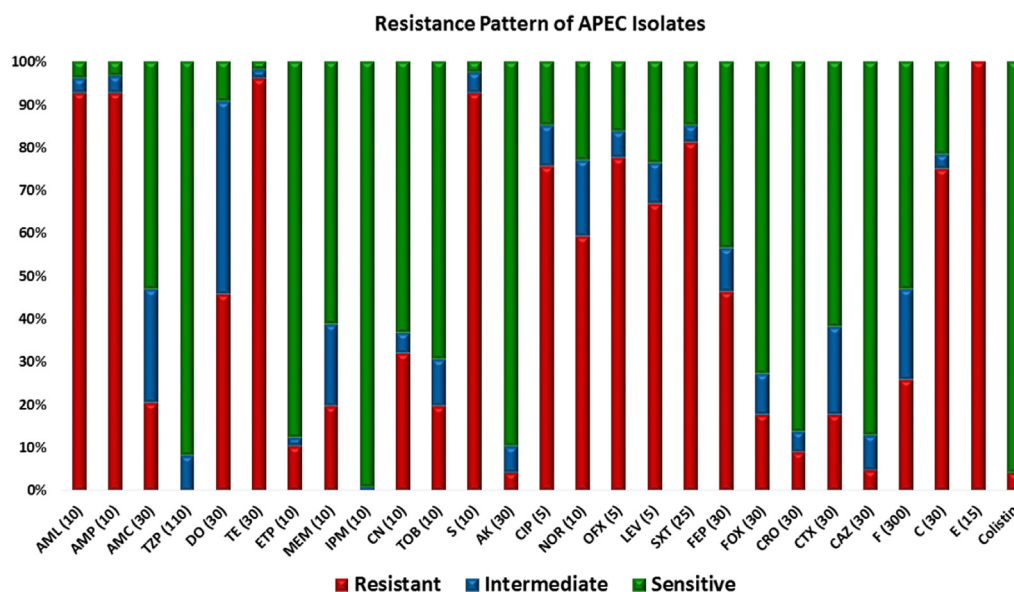


Fig. 2. In vitro antibiotic susceptibility pattern of 147 APEC isolates; stacked bar chart showing the percentage resistance and sensitivity of APEC isolates towards 27 tested antibiotics. The antibiotics used in AST profiling included (1) amoxicillin (AML, 10µg), (2) Ampicillin (AMP, 10µg), (3) amoxicillin-clavulanate (AMC, 20/10µg), (4) doxycycline (DO, 30µg), (5) tetracycline doxycycline (TE, 30µg), (6) piperacillin-tazobactam (TZP, 100/10µg), (7) erapipenem (ETP, 10µg), (8) meropenem (MEM, 10µg), (9) imipenem (IPM, 10µg), (10) gentamicin (CN, 10µg), (11) tobramycin (TOB, 10µg), (12) streptomycin (S, 10µg), (13) amikacin (AK, 30µg), (14) ciprofloxacin (CIP, 5µg), (15) norfloxacin (NOR, 10µg), (16) ofloxacin (OFX, 5µg), (17) levofloxacin (LEV, 5µg), (18) trimethoprim-sulfamethoxazole (SXT, 25µg), (19) cefepime (FEP, 30µg), (20) ceftazidime (CAZ, 30µg), (21) ceftriaxone (CRO, 30µg), (22) cefotaxime (CTX, 30µg), (23) ceftazidime (CAZ, 30µg), (24) nitrofurantoin (F, 300µg), (25) chloramphenicol (C, 30µg), (26) erythromycin (E, 15µg), (27) colistin sulfate.

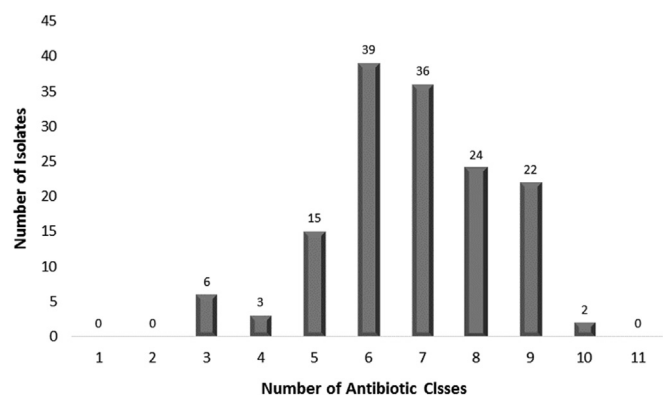


Fig. 4. The bar chart shows the number of APEC isolates resistant to 12 tested antibiotic classes. The x-axis shows the number of antibiotic classes, while the y-axis shows the number of *Escherichia coli* isolates resistant to specific numbers of antibiotic classes. The antibiotic classes that were used for antibiotic susceptibility profiling include: (I) penicillin, (II) tetracycline, (III) penicillin β -lactamase inhibitor, (IV) carbapenem, (V) fluoroquinolone, (VI) aminoglycoside, (VII) cephalosporin, (VIII) nitrofurantoin, (IX) phenicol, (X) macrolide, (XI) colistin, and (XII) sulfonamide.

3.5. Detection of antibiotic resistance genes in the APEC isolates

Overall, 81 isolates (55.1%) were positive for *bla*TEM, while *bla*CTX was detected in 49 isolates (33.3%). Twenty-seven isolates (18.4%) were positive for both *bla*TEM and *bla*CTX genes. All APEC isolates were negative for *bla*SHV, *bla*NDM, *bla*IMP, *bla*VIM, and *bla*KPC (Fig. 3). Furthermore, 59 APEC isolates (40.1%) were positive for *mcr*-1 and all isolates were negative for *mcr*-2 gene (Fig. 3).

3.6. In vitro pathogenicity analysis of APEC isolates

The biofilm formation assay confirmed that 48 isolates (32.65%) were unable to form biofilm and, 39 APEC isolates (26.53%) exhibited weak biofilm formation, 40 isolates (27.21%) formed moderate biofilms, and 20 isolates (13.6%) exhibited strong biofilm formation (Fig. 7A, 7C). Strong biofilm formation was most prominent in phylogroups B2 and D (Fig. 7B). Growth in human urine was evident in 99 (67.4%) of the 147 tested APEC isolates (Fig. 7D). For congo red binding assay, a total of 129 isolates (87.8%) were able to bind to the congo red dye (demonstrated by the red appear-

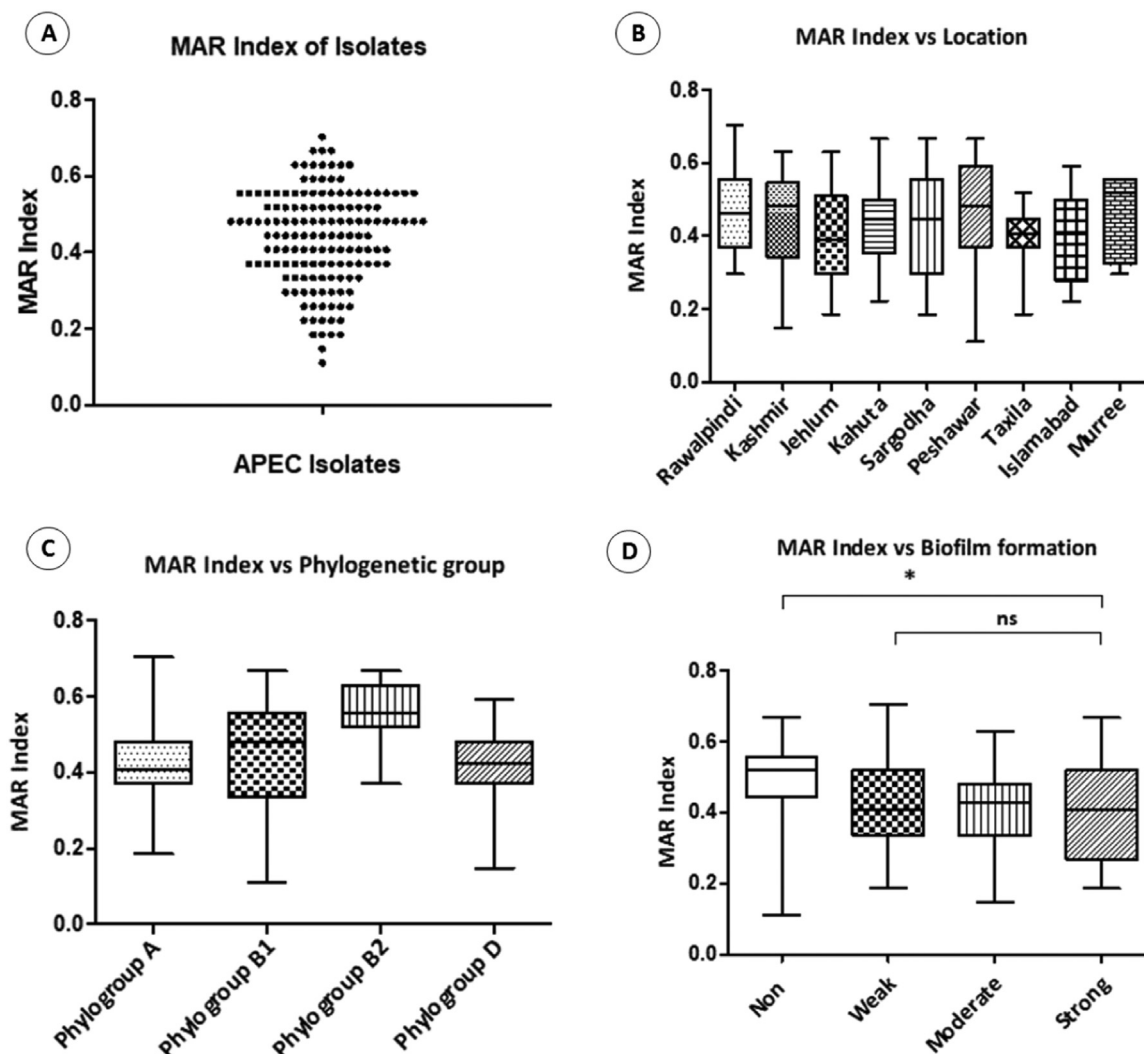


Fig. 5. Phenotypic associations: (A) scatter plot showing the MAR index of APEC isolates; dots represent each isolate's response to show variability and distribution of data (B) box plot showing the comparison of the MAR indices of various locations from where the APEC strains were isolated; the means were generally different for each location but statistically they were non-significant ($P = 0.5297$) (C) boxplots showing the MAR indices of isolates from different phylogenetic groups; the means were generally significantly different ($P = 0.084$) (D) boxplots showing the MAR indices of isolates from different biofilm forming groups; the means were generally significantly different ($P = 0.003$).

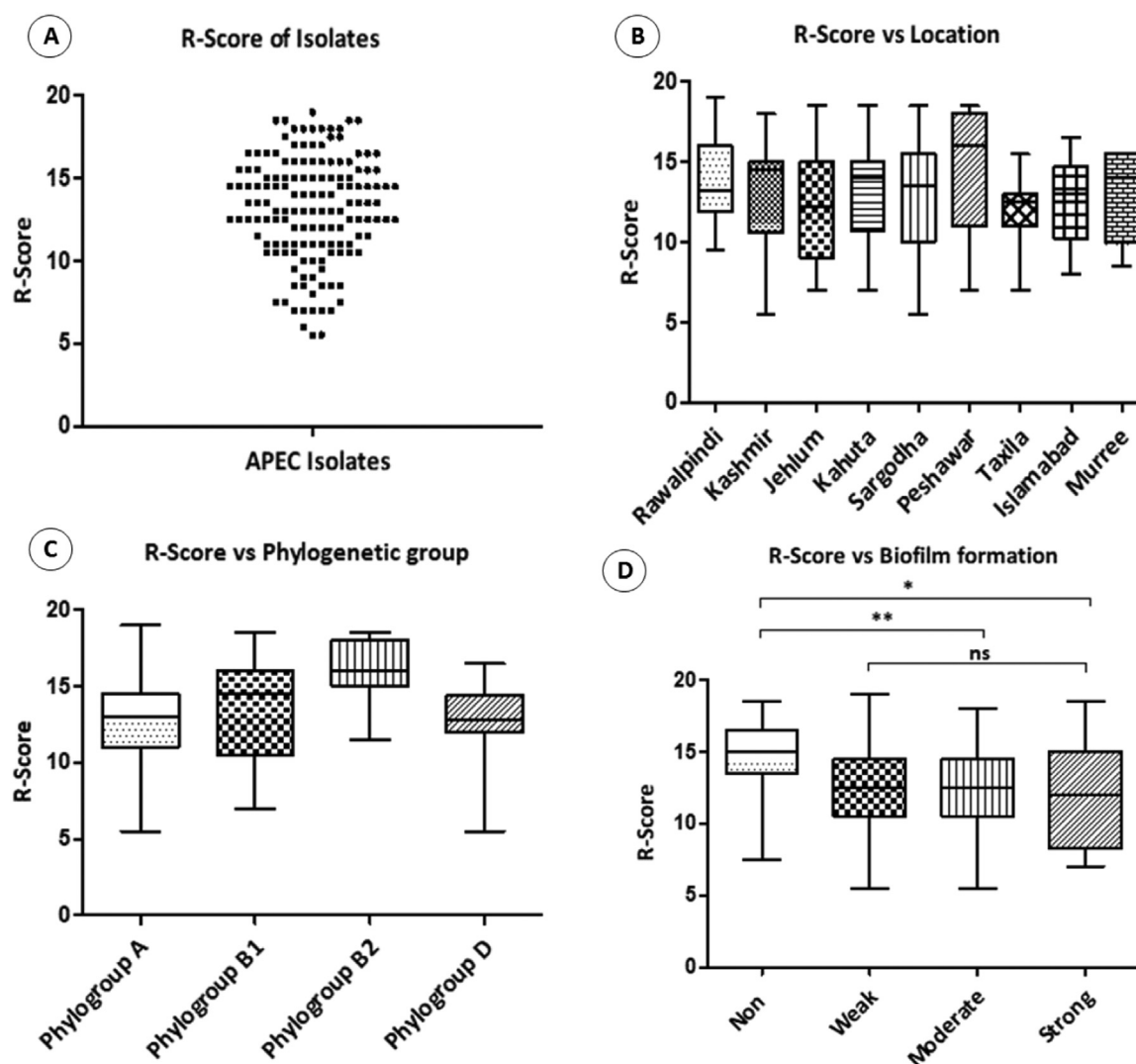


Fig. 6. Phenotypic associations: (A) scatter plot showing the R-scores of APEC isolates; dots represent each isolate's response to show variability and distribution of data (B) box plot showing the comparison of R-scores of various locations from where the APEC strains were isolated; the means were generally different for each location but, statistically, they were non-significant ($P = 0.6392$) (C) boxplots showing R-scores of isolates from different phylogenetic groups; the means were generally significantly different ($P = 0.0723$) (D) boxplots showing R-scores of isolates from different biofilm forming groups; the means were generally significantly different ($P = 0.0004$).

ance of APEC colonies on CRA streaked plates), indicating their potentially invasive nature; 18 isolates (12.2%) remained unbound to congo red dye (demonstrated by greyish/pinkish growth). All APEC isolates showed γ -hemolysis. Overall, 56 (38.1%) of the 147 tested APEC isolates were complement resistant, as their OD_{492} in serum-containing wells exceeded or equalled that of the no-serum control wells after 4 h of growth. All APEC isolates were negative for lipase and protease activity.

As far as swimming and swarming motility are concerned, APEC isolates were considered motile if they displayed a halo zone above 10 mm. Overall, 102 (69.4%) of the 147 APEC isolates showed swimming motility (Fig. 8A, 8B). For swarming motility, only 33 isolates (22.4%) of 147 tested isolates were motile (Fig. 8C, 8D). For twitching motility, isolates were considered motile if they displayed a zone size ≥ 6 mm. Overall, 86 (58.5%) of the 147 APEC isolates exhibited twitching motility (Fig. 8E, 8F).

3.7. Phylogenetic classification of APEC isolates

The APEC isolates were phylogenetically classified into four groups: A, B1, B2, and D. Results exhibited that the most preva-

lent phylogenetic group was B1 (51%), followed by A (30.6%), D (13.61%), and B2 (4.76%) (Fig. 9).

3.8. Prevalence of VAGs

Virulence genotyping of 147 APEC isolates identified 20 of the 24 studied VAGs in at least one isolate each. The four exceptions were genes for certain adhesins (*ibeA*, *bmaE*, *afa*, and *dra*). Among the 20 detected genes, prevalence values ranged from 0.7% to 91.8%. At a very low prevalence (1–10%) were certain adhesins: *sfa* (0.7%), *foc* (0.7%), *papA* (5.4%), *papEF* (5.4%), *papG* (5.4%), and *papC* (6.1%); toxins: *hlyA* (0.7%) and *cnf1* (0.7%); protectins: *KpsMT (K1)* (4.8%), *kpsMT III* (5.4%), and *KpsMT II* (6.1%); and siderophore *ireA* (9.5%). At a low prevalence (11–30%) were siderophore *fyuA* (12.2%) and *iutA* (29.2%), autotransporter *tsh* (12.2%), and adhesin *cvaC* (21.1%). At a medium prevalence (31–60%) were siderophore *iroN* (43.5%), and protectins *traT* (51.7%) and *iss* (56.5%). Finally, at the highest prevalence (80–100%) was adhesin *fimH* (91.8%) (Fig. 10).

Distribution of VAGs subgroups was significantly different in different phylogenetic groups. The highest number of VAGs, along

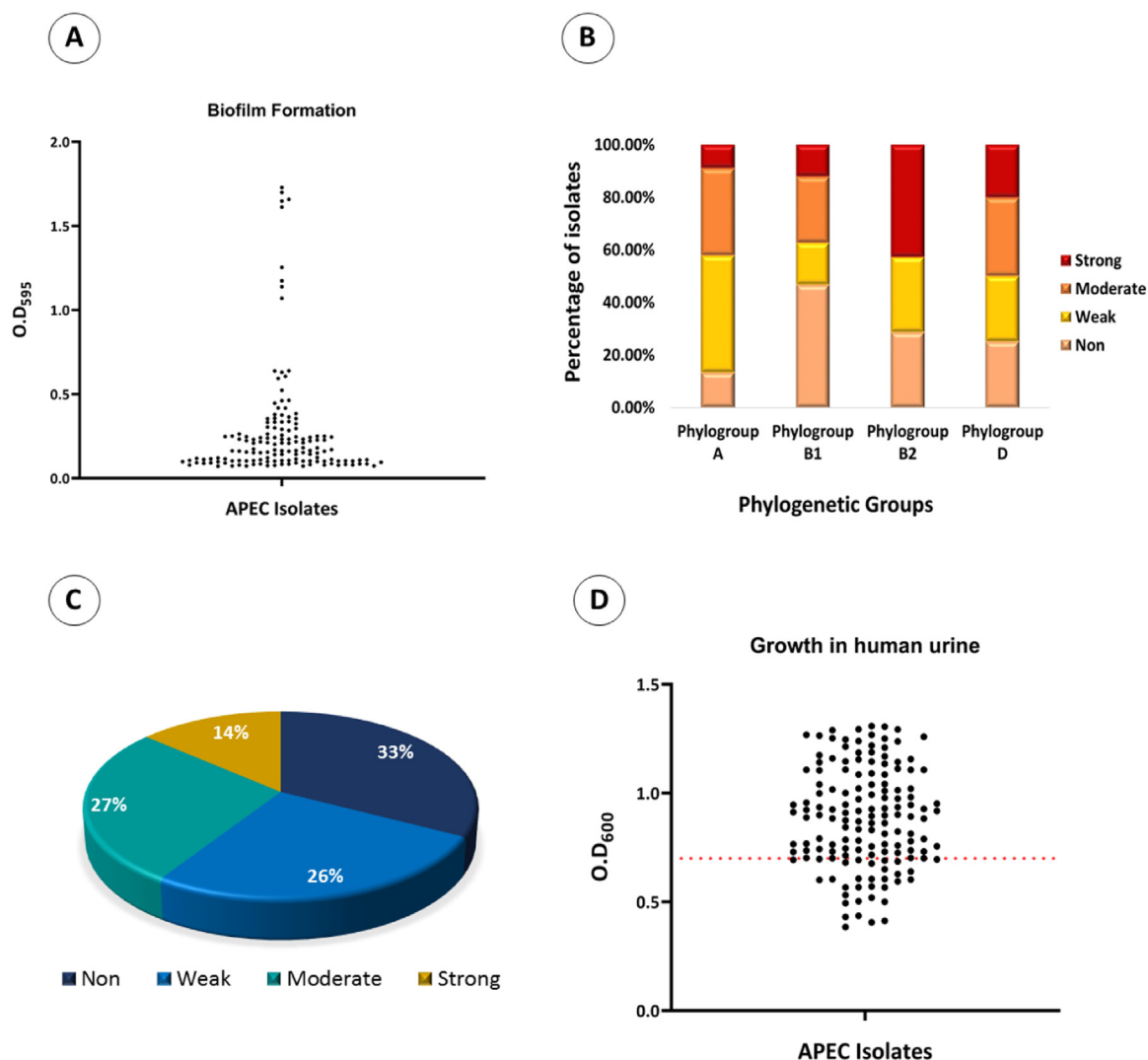


Fig. 7. (A) scatter plot showing the biofilm formation by APEC isolates; dots represent each isolate's biofilm value (B) stacked bar chart showing the comparison of biofilm formation by *Escherichia coli* strains from different phylogenetic groups (C) pie chart showing the percentage biofilm formation by APEC isolates (D) scatter plot showing the growth in human urine by APEC isolates; dots represent each isolate's optical density value and the red line is representing the standard optical density (OD) value required by the strains to be called positive for growth in human urine; the strains at or above the red lines are positive for growth in human urine.

with all its subclasses, was observed in phylogroup B, followed by phylogroup D (Fig. 11).

3.9. Frequency of hybrid strains

Among the 147 APEC isolates evaluated, 26 (17.69%) presented at least one of the key markers used to define the DEC pathotype; thus, they were considered as hybrid strains. Twenty-five strains (17%) were considered as hybrid APEC/EHEC pathotypes, as they carried the *stx1* gene. Two strains (1.36%) harboured *eaeA* but not *bfpB*, and were considered as hybrid APEC/aEPECs (atypical EPECs); one of these hybrid strains also carried *stx1* and was considered as hybrid APEC/aEPEC/EHEC. No ETEC, EIEC, or EAEC virulence markers were identified among the 147 APEC strains (Fig. 12).

3.10. ERIC-PCR fingerprints analysis for evaluation genetic diversity of APEC isolates

The ERIC-PCR analysis showed that APEC isolates produced 1 to 11 fragments per isolate, ranging in size from 100 to 2000 bp. Overall, 147 isolates were grouped into 40 clusters (E1–E40) at 80%

genetic similarity with each group containing at least two isolates, and only 30 isolates remained unclustered (Fig. 12).

4. Discussion

It is speculated that, by 2050, *E. coli* will transform into incurable bacterial strains as they continue to develop resistance against many known antibiotics [38]. Generally, most of the *E. coli* are natural inhabitants of the mammalian intestine; however, after acquiring certain virulence-associated genes, they can cause intestinal and extra-intestinal infections that result in high morbidity and mortality around the globe [39]. In developing countries like Pakistan, antibiotics are extensively used in the poultry industry for infection control and as growth promoters. This continuous exposure to antibiotics employs selection on bacteria and is enhanced by their ability to acquire different resistance factors from other surrounding bacteria, leading to the emergence of modified strains with diverse resistance traits [40]. Despite these concerns, global antimicrobial consumption by food animals is expected to increase by 67% between 2010 and 2030 [41]. This study aimed to determine the virulence potential, antibiotic resistance

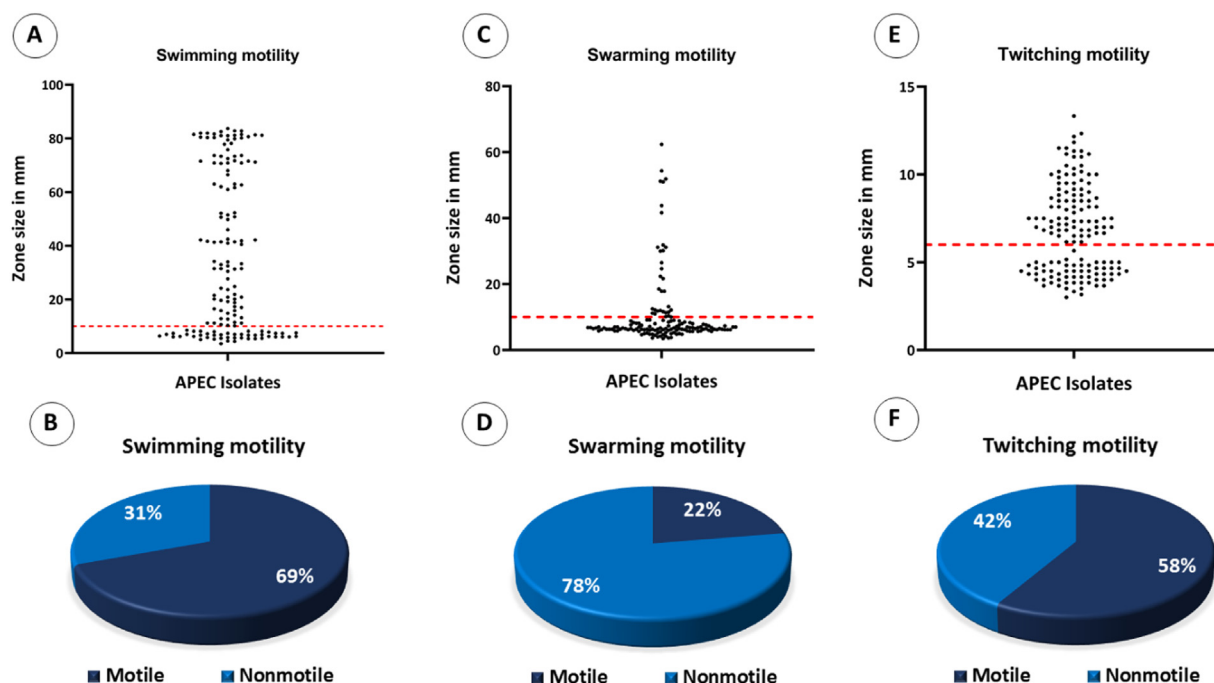


Fig. 8. Motility assays by APEC isolates; dots in scatter plots represent each isolate's motility in respective media and the red line represents the standard cut-off value required by the strains to be called motile; the strains at or above the red lines are motile. (A) Scatter plot showing swimming motility by APEC isolates (B) pie chart showing the percentage of isolates positive and negative for swimming motility (C) scatter plot showing swarming motility by APEC isolates (D) pie chart showing the percentage of isolates positive and negative for swarming motility (E) Scatter plot showing twitching motility by APEC isolates (F) pie chart showing the percentage of isolates positive and negative for twitching motility.

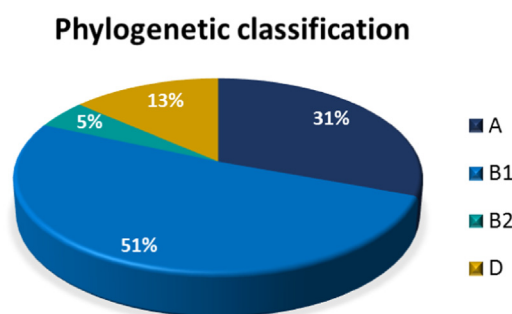


Fig. 9. Pie chart showing the relative distribution of phylogenetic groups among APEC isolates.

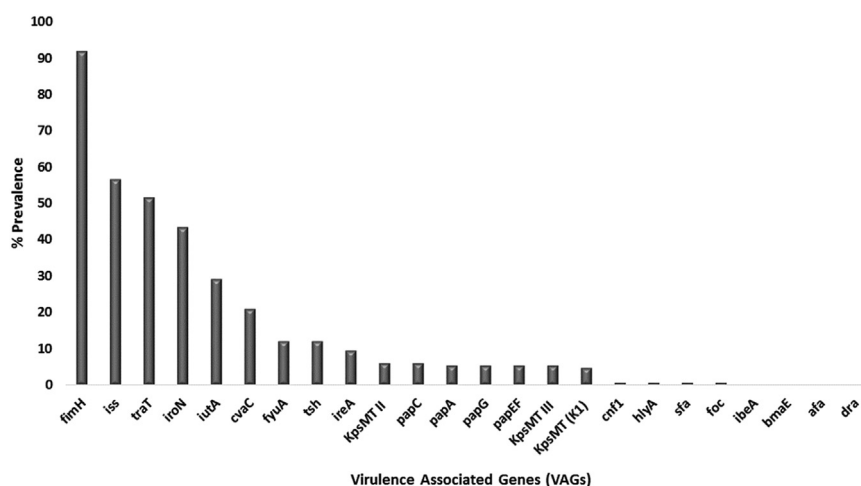


Fig. 10. Bar graph showing the percentage prevalence of virulence associated genes (VAGs) in APEC isolates.

profile, and frequency of hybrid strains in APEC strains isolated from colibacillosis-infected livers in several regions of Pakistan.

Resistance patterns of bacterial isolates are generally dependent on their geographical locations and seem to be associated with frequent use of certain antibiotics in different regions. In this study, the highest resistance was found against the penicillin, tetracycline, macrolide, and fluoroquinolone classes of antibiotics, implying overuse of these antibiotics in local poultry farms to treat infections. One of our recent studies involving *E. coli* isolates from healthy cows also showed resistance against these antibiotic classes [34]. Fluoroquinolones are broad-spectrum synthetic drugs that are used extensively to control and treat bacterial infections worldwide. Resistance to this antibiotic class has emerged because of its widespread use in the poultry industry

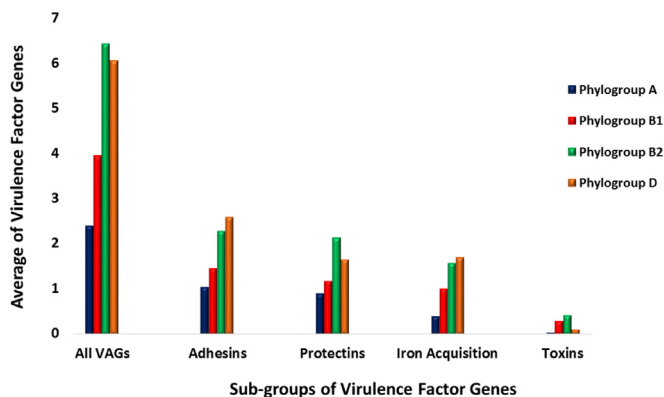


Fig. 11. Bar graph showing the virulence factor gene subgroups amongst most phylogenetic groups of APEC isolates. Bars represent the average values of the genes present in that phylogroup.

and likely increases the probability of treatment failures [42]. Furthermore, the prevalence and dissemination of resistance to fluoroquinolones in APEC have significantly increased in recent years because fluoroquinolone-resistant *E. coli* isolates often exhibit MDR phenotypes [43–46]. In this study, fluoroquinolone-resistant APEC isolates showed co-resistance to penicillin and tetracycline; these results are consistent with those of recent studies [43–47].

The antibiotic resistance of the studied APEC isolates was very high, as all isolates were MDR (they were resistant to at least one agent in three or more antibiotic classes) [48]. In this study, the prevalence of MDR *E. coli* isolates (100%) was higher than what has been previously reported worldwide: 8% in dairy farms of Canada [49], 13.7% in South Africa [50], 37.1% in Jordan [51], 44.4% in Egypt [52], 56.0% in France [53], and 69.0% in Portugal [54], and 95.6% in poultry farms of China [55]. An increase in resistance to one antibiotic class has been suggested to increase the chances of recurrent infection by 1.6 (i.e., isolates harbouring resistance to five classes of antibiotics have 1.6 times more chances of causing recurrent infection than the isolates resistant to four antibiotic classes) [56]. Therefore, thoughtful use of antibiotics is strictly recommended because overuse further drives bacterial populations towards resistance. One of the great concerns was the presence of the mobilized colistin gene, *mcr-1*, in 40% of the isolates, though only 4% were showing phenotypic resistance. Since its discovery in swine *E. coli* isolated from China, it has been found in various *E. coli* isolates, including a collection of avian *E. coli* from Egypt and China [57,58]. Some studies from Pakistan found an *mcr-1* prevalence of 8% in APEC isolates from Faisalabad [59] and 14% in APEC isolates from various regions of Pakistan [60].

The increase in resistance to major antibiotics might be due to the acquisition of β -lactam-inactivating enzymes called β -lactamases, which include ESBLs, metallo- β -lactamases (MBLs), AmpC cephalosporinases, and carbapenemases found in bacterial strains worldwide [61]. This study supports this hypothesis, as more than 85.7% of APEC isolates in this study were found to harbour ESBL genes; the situation is problematic, as these genes are plasmid-borne and can be transferred to other bacteria, environments, and humans [62].

Subsets of *E. coli* from animal sources are known to cause various intestinal and extraintestinal pathologies in humans. It has been suggested that APEC isolates with higher biofilm production and motility are more pathogenic and resistant to antimicrobials and host defences [63,64]. Biofilms can help in the acquisition of genetic material by horizontal gene transfer, as they allow bacteria to be near other pathogens [65]. In this study, 99 of 147 isolates (67.3%) demonstrated biofilm formation. Moreover, complement resistance enhances the ability of extraintestinal infections to avoid

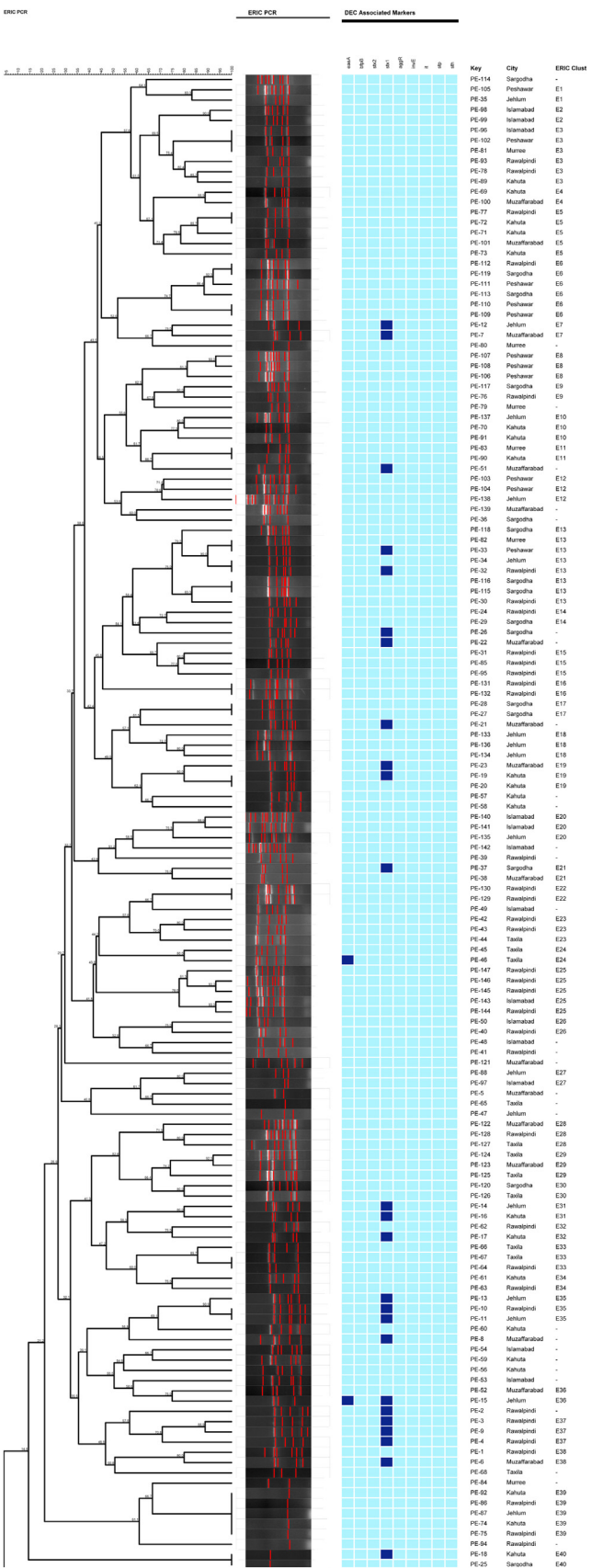


Fig. 12. The Enterobacteria repetitive intergenic consensus polymerase chain reaction (ERIC-PCR) analysis of the APEC isolates from colibacillosis-infected liver samples is displayed using dendrograms generated by Bionumerics software. The delineation level is set at 80%. The heatmap depicts the presence/absence of genes: dark blue represents the presence, while light blue represents the absence, of diarrheagenic *E. coli* (DEC)-associated markers.

immune clearance [66]. Most APEC isolates were able to grow in human urine, showing that the APEC isolates had the potential to survive in high-urea and nitrogenous environments.

Escherichia coli strains were classified into phylogenetic groups A, B1, B2, and D; the virulent ExPEC strains belonged mostly to B2 and D [37]. This was in accordance, as a high prevalence of VAGs, as well as its subgroups, was observed in phylogroups B2 and D. The high prevalence of *fimH* genes (91.8%) in the APEC isolates indicated their ability to adhere to or colonize extraintestinal environments. Together, with the high prevalence of *fimH* and high potential for bacteriuria, these isolates might be zoonotic. *fimH* codes for an adhesin gene that facilitates the binding of isolates with α -mannoside present on the epithelial lining of the upper urinary tract and various human organs [67]. The high prevalence of *iss* and *traT* indicates that many of these isolates might survive in serum by resisting complement-mediated killing. The prevalence of *iss* is much higher in ExPEC, though it is also expressed in commensal bacteria [68].

Several studies suggest that *E. coli* is continuously evolving because of genome deletions and horizontal gene transfer, leading to the emergence of complex strains with embellished pathogenic potential and significant intra-species variability [69,70]. The most prominent example is the emergence of hybrid-pathogenic *E. coli* that carry combinations of ExPEC and DEC VFs. Some studies have detected DEC VFs among ExPEC isolates, but more studies are required to understand their frequency and effect on animal health and clinical outcomes [70,71]. Overall, the frequency of hybrid *E. coli* strains detected in APEC samples was 26 (17.69%).

Molecular typing of *E. coli* is vital for determining potential sources of infection and implementing effective epidemiological surveillance and control. Bacterial genomes containing ERIC sequences (repeat sequences) can be used as tools to evaluate the clonal variability of bacterial isolates [72]. ERIC-PCR fingerprinting is one of the fastest and most reliable molecular typing techniques used to differentiate between *E. coli* and other Gram-negative bacterial strains [73]. In this study, 117 of 147 antimicrobial-resistant APEC isolates were grouped into nine ERIC-PCR clusters, and 30 isolates remained unclustered. This genetic variability might be due to differences between the sources of the samples.

In conclusion, this study illustrates the current state of resistance to fluoroquinolone and other antibiotics in APEC isolates of the investigated poultry farms, and the finding that 100% of the *E. coli* isolates were MDR is alarming. The high prevalence of *bla*CTX and *bla*TEM in *E. coli* was detected and is of high concern, as the co-occurrence of different types of β -lactamases in *E. coli* presents challenges in both diagnosis and treatment. The growth of APEC isolates in human urine and serum indicates their zoonotic potential. This study contributes to a better understanding of the frequency, occurrence, diversity, and pathogenicity of hybrid *E. coli* strains, which can be involved in severe cases of colibacillosis and cause other intestinal and extraintestinal diseases. The findings of this study provide insight into the antibiotic resistance and virulence profiling of APEC isolates in Pakistan. This data can inform future studies designed to better estimate the severity of colibacillosis in poultry farms.

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Ethical approval

All protocols, including sampling, handling of colibacillosis-infected liver samples, transportation of samples, storage of sam-

ples, and analysis of the results were approved by the Board of Advanced Studies and Research (BASR) of the National University of Sciences and Technology (NUST), Islamabad. In addition, written consent was taken from the farm handlers before sampling.

Institutional review board statement

All the analysis of this study was approved by the Institutional Review Board (IRB) committee of the Industrial Biotechnology Department of NUST (Ref No: IRB-88).

Competing interests

None declared

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.jgar.2023.01.009](https://doi.org/10.1016/j.jgar.2023.01.009).

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